

****Publication of Dataset and AI Model****

Date: 19-01-2024

****Decision****

Public Authorities

Response to Inquiry

Legal Basis for Processing

Fundamental Principles

Sønderborg Municipality has requested the Data Protection Agency's assessment of whether the municipality can publish a dataset that it has established and used for the development of an AI model, as well as whether the municipality can publish the developed model within the framework of data protection regulations.

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****Summary****

Sønderborg Municipality has requested the Data Protection Agency's assessment of whether the municipality can publish a dataset that it has established and used for the development of an AI model, as well as whether the municipality can publish the developed model within the framework of data protection regulations.

The Data Protection Agency has assessed that the data protection regulations do not prevent the basic model from being published.

Regarding the publication of the dataset, the prerequisite for the dataset to be lawfully published is that the personal data included in the dataset has been lawfully collected and processed.

The Data Protection Agency's assessment is that Sønderborg Municipality has lawfully been able to collect and process the relevant information, and that the municipality can lawfully publish the dataset. However, the municipality must be aware of the specific requirements regarding data minimization, accuracy, and legality as part of its potential publication.

You can read the Data Protection Agency's statement to Sønderborg Municipality below.

1. ****The Inquiry****

Sønderborg Municipality requested on June 8, 2023, the Data Protection Agency's assessment of whether the municipality can publish a dataset that it has established and used for the development of an AI model, as well as whether the municipality can publish the developed model within the framework of data protection regulations.

2. ****The Data Protection Agency's Assessment****

It is – after the case has been presented to the Data Council – the Data Protection Agency's assessment that the developed AI model does not in itself constitute personal data, and that the exhibition of the basic model is therefore, as a starting point, not covered by the data protection regulations.

This means that the data protection regulations do not prevent the basic model from being published.

Regarding the publication of the dataset, the prerequisite for the dataset to be lawfully published is that the personal data included in the dataset has been lawfully collected and processed.

In this context, the Data Protection Agency's assessment is that the collection and processing of personal data as part of the establishment of a dataset for the development of an AI solution, which is to support the municipal processing of access to information cases, can occur with reference to the access to information provisions in the Public Access to Information and Administrative Act, cf. Article 6, paragraph 1, letter e, and Article 9, paragraph 2, letter g, of the General Data Protection Regulation in conjunction with Article 6, paragraphs 2 and 3.

Furthermore, it is the Data Protection Agency's assessment that the dataset can be published on sprogteknologi.dk with reference to Article 6, paragraph 1, letter e, and Article 9, paragraph 2, letter g, of the General Data Protection Regulation, with reference to the Public Access to Information and Administrative Act as a supplementary legal basis.

The Data Protection Agency notes, however, that the municipality, as part of the publication, must be aware of the requirements regarding data minimization, cf. Article 5, paragraph 1, letter c, the accuracy

of the information, cf. Article 5, paragraph 1, letter d, as well as the legality of the information, cf. Article 5, paragraph 1, letter a, as further described in section 3.4.2.

3. ****Background for the Assessment****

3.1. ****About the Project****

Sønderborg Municipality is part of an inter-municipal collaboration on the development and use of an AI tool that is to support the municipal processing of access to information cases.

The purpose of the collaboration is to develop language models that can identify information in case documents that may need to be extracted as part of responding to access to information requests. This may include, for example, information about personal identification numbers or other confidential information.

Once the language models are developed, they will be integrated into an IT solution used for processing access to information requests. The language models will thus be used as decision support for caseworkers. The models will, for example, be able to identify names across a large number of documents and suggest that these be extracted. However, it is ultimately the caseworker who decides whether the information should be extracted.

In the inter-municipal collaboration, three types of models have been developed – two complex models and one basic model – as well as a dataset.

The dataset has been created based on publicly available data from Danish municipalities' websites (data scraping). It therefore includes, for example, names of municipal employees and municipal politicians, job titles, addresses, emails, phone numbers, and information appearing in minutes from city council and committee meetings, etc.

As part of the project, Sønderborg Municipality wishes to make both the basic model and the dataset publicly available on sprogteknologi.dk. The background for sprogteknologi.dk includes the following[1]:

"The digital platform sprogteknologi.dk collects metadata about Danish language resources in one place. The primary purpose is to support the development of artificial intelligence in Danish and thereby contribute to...

ensure that the digital language in Denmark is Danish. The platform sprogteknologi.dk contains, among other things, metadata about speech and text corpora, dictionaries, term bases, language technology support tools, and infrastructure components. The digital platform will continuously be expanded with new metadata about relevant existing language resources. In the long term, new language resources will also be developed and showcased, which can reduce [barriers] and strengthen the development of high-quality Danish language technology solutions to support artificial intelligence."

From Sønderborg Municipality's impact assessment of the tool, it appears that the processing of personal data for the development of the tool is based on Article 6, paragraph 1, letter e of the General Data Protection Regulation. Special categories of personal data are processed based on the exceptions in Article 9, paragraph 2, letters a, b, c, f, g, h, and j, while information about criminal offenses is processed based on Section 8, paragraph 1 of the Data Protection Act. Information about personal identification numbers is processed based on Section 11, paragraph 1 of the Data Protection Act. The impact assessment is publicly available.[2]

3.2. **Sharing the Base Model**

The question is primarily to what extent Sønderborg Municipality can make the base model publicly available. According to the Danish Data Protection Agency's guidance on the use of artificial intelligence by public authorities (Before You Start) from October 2023, the following is stated: "The Data Protection Agency assumes that an AI model, as a clear starting point, does not in itself constitute personal data, but is merely the result of the processing of personal data. This is similar to a statistical report not being considered personal data if the report only contains conclusions and aggregated data that are the results of the statistical analysis.

However, certain machine learning models can be attacked in various ways (so-called model inversion attacks and membership inference attacks), which make it possible to re-identify the citizens whose

data has been included in the model's training data. A successful attack that results in the re-identification of citizens' data in the training data may constitute a breach of data security and must be handled accordingly. The risk that a malicious actor re-identifies citizens by deliberately conducting an attack to infer data that has been included in the training data does not, in the opinion of the Data Protection Agency, mean that the model should be considered personal data in itself."

This view is primarily based on the understanding that AI solutions constitute systems that, by using training data, are trained to identify certain patterns, after which the solution can identify the same patterns in new data. Thus, the systems do not contain information in themselves, unlike actual databases, but only contain the logic and relationships in the patterns that the solution has been trained to recognize.

On this basis, it is the Data Protection Agency's assessment that the exhibition of the base model is, as a starting point, not covered by data protection regulations. However, the Data Protection Agency notes that Sønderborg Municipality must be aware of the aforementioned risk of attacks aimed at re-identifying citizens whose data has been included in the training data. If the municipality becomes aware of such attacks, it should take appropriate steps to secure the model or remove it from the internet.

3.3. Establishment of the Dataset

The prerequisite for the dataset to be legally published on sprogteknologi.dk is that the personal data included in the dataset has been lawfully collected and processed. Therefore, the inquiry gives rise to consider whether the personal data has been lawfully collected and processed.

3.3.1. General Data Protection Regulation

In the opinion of the Data Protection Agency, it is primarily Article 6, paragraph 1, letter e of the General Data Protection Regulation regarding the exercise of public authority that is a relevant basis for the collection and processing of personal data as part of the establishment of the dataset.

Processing of personal data based on Article 6, paragraph 1, letter e of the General Data Protection Regulation requires that there is a so-called supplementary legal basis that obligates or authorizes the authority to perform a specific public task. This follows from Article 6, paragraphs 2 and 3 of the Regulation.

Furthermore, it is the Data Protection Agency's opinion that the prohibition against processing special categories of personal data in Article 9, paragraph 1 of the General Data Protection Regulation may be deviated from under Article 9, paragraph 2, letter g of the Regulation. This provision states that the prohibition in paragraph 1 does not apply if processing is necessary for reasons of substantial public interest based on EU law or national law of the member states and is proportionate to the aim pursued, respects the essential content of the right to data protection, and ensures appropriate and specific measures to protect the fundamental rights and interests of the data subject.

A deviation from the prohibition in Article 9, paragraph 1 of the General Data Protection Regulation under Article 9, paragraph 2, letter g, requires, as is the case with Article 6, paragraph 1, letter e, that there is a supplementary legal basis. The Data Protection Agency has, in its statement of November 17, 2023, to the Municipality of Copenhagen, elaborated on the requirements for the supplementary legal basis for the processing of personal data as part of the development, operation, and retraining of AI solutions.[3] Reference is made to section 3.2 of the statement.

3.3.2. Public Access and Administration Act

Sønderborg Municipality has, in its impact analysis, broadly referred to the fact that the municipality, as a public authority, is obligated to process personal data as part of handling requests for access to documents under the Public Access and Administration Act.

According to the Data Protection Authority, there is no requirement that a legal basis must be found in Articles 6 or 9 of the General Data Protection Regulation (GDPR) in order to disclose information in connection with responding to requests for access to documents, nor is there, among other things, an (additional) requirement for data minimization, as the issue of access to documents is solely regulated by the rules of the Administration Act, the Public Access Act, or the Environmental Information Act.

The legislator has assumed that the rules on access to documents in these laws are compatible with the Regulation, cf. its Article 86.

Establishing a dataset consisting of publicly available information for the purpose of developing an AI solution to support the processing of requests for access to documents must, however, according to the Data Protection Authority, be considered a distinct and separate purpose from the actual processing of specific requests for access to documents.

Reference is made to the Data Protection Authority's guidance on the use of artificial intelligence by public authorities (Before You Start), p. 9, which states:

"The development of an AI solution should, in the opinion of the Data Protection Authority, be regarded as a purpose in itself in the context of data protection rules. Processing of personal data for the purpose of developing new technological solutions serves, by its nature, a different purpose than processing of personal data as part of the authority's daily operations, e.g., as part of the municipality's case processing or the region's health initiatives towards specific citizens. This also applies even if the long-term purpose of developing the solution is to use it in the authority's daily operations."

While the assessment of whether there has been an unjustified disclosure of personal data as part of granting access to documents must be made based on a "pure" application of the rules in the Public Access and Administration Act, it can be assumed that the Public Access and Administration Act may constitute a supplementary national legal basis under Articles 6(2) and (3) of the GDPR for the processing of personal data as part of the development of, for example, AI solutions.

In this context, the Data Protection Authority holds the view that the collection and processing of personal data as part of establishing a dataset for the development of an AI solution, which is to support the municipal processing of access to documents, can take place with reference to the access to documents provisions in the Public Access and Administration Act, cf. Article 6(1)(e) of the GDPR, and Article 9(2)(g), in conjunction with Articles 6(2) and (3).

The Data Protection Authority has particularly emphasized that the establishment of the dataset and the development of the solution occur in direct connection with the tasks that the municipality is obligated to perform, particularly the granting of access to documents. Furthermore, the processing of personal data in question does not have direct consequences for citizens, as it concerns the development of a solution.

3.4. Sharing of Datasets

Finally, the question arises as to the extent to which Sønderborg Municipality can share the established dataset on sprogteknologi.dk so that other authorities can use the dataset to develop similar solutions.

3.4.1. Legal Basis for Processing

According to the Data Protection Authority, it is also Article 6(1)(e) of the GDPR, as well as possibly the exception in Article 9(2)(g) of the Regulation, that is the relevant legal basis for the municipality's sharing or publication of the dataset.

Sønderborg Municipality has stated that it wishes to make the dataset publicly available on sprogteknologi.dk. The Data Protection Authority therefore assumes that the purpose of the publication is more specifically to contribute to and support, among other things, the development of AI solutions in Danish by other public authorities.

It is generally recognized that public authorities—like private companies—can also develop new technological solutions to support their exercise of authority.

It is thus stated, among other things, in Section 1(2) of the Public Access Act, that authorities "must ensure that the consideration for openness mentioned in subsection 1 is safeguarded to the greatest extent possible when choosing, establishing, and developing new IT solutions."

Regarding the background for this provision, the following is stated in Chapter 11, Section 4.5 of the Public Access Commission's report No. 1510/2009, Volume I:

"As indicated in point 1 above, technological development has led to public administration increasingly using various types of databases as a basis for their operations, including in connection with administrative case processing.

It is the Commission's opinion that the continued technological development is expected to entail that administrative authorities will use new types of digital tools in connection with administrative case

processing. It is therefore the Commission's opinion that public administration, when choosing technology, should take into account the purposes that the Public Access Act aims to support, cf. the draft law's Section 1(1), nos. 1-5.

Administrative authorities should therefore, when establishing and developing new digital tools, keep in mind that they are designed in such a way that they are

****Foreign to the Realization of the Principle of Public Access****

including the possibility that access to documents may, if applicable, be granted via email.

Administrative authorities should also, in the establishment and development of (new) databases, keep in mind that each database should be designed in such a way that it is possible, through simple commands, to extract data in accordance with the draft law's § 11. [...]”

In the administrative law literature, the same assessment is made. It is stated, among other things, in “From Administrative Lawyer to Development Lawyer,” Hanne Marie Motzfeldt et al., 1st ed., 2020, p. 59: “As far as the requirement for legal authority is concerned, it cannot, as one extreme, be required that there is a legal basis in formal law to purchase and use simple tools such as word processing systems, journal systems, etc. Establishing budgetary authority is sufficient. As the other extreme, extensive digitization projects can lead to significant changes in the organization of the administration, governance and accountability relationships, and workflows, as well as for the 'meeting' with citizens. If there is such extensive impact on the way the administration functions, ideal and principled considerations regarding the subordinate position of the administration in the state governance suggest that the democratically legitimized legislator should be involved.”[5]

The Data Protection Agency has also recognized through its practice that public authorities – within the framework of their exercise of authority – can develop new IT tools and process personal data as part of the development. Reference can be made to the Data Protection Agency's guiding text on the use of personal data for testing purposes, where it is assumed that, among others, authorities can process personal data as part of development work.[6]

According to the Data Protection Agency's practice, there is no basis for deviating from this starting point regarding Sønderborg Municipality's sharing of the dataset.

In the opinion of the Data Protection Agency, it can be said to fall within the framework of ordinary development work to share, for example, datasets such as the one in question, especially in light of the fact that the development of AI solutions largely relies on the use of (multiple) datasets, and that the quality of the developed solutions largely depends on the data used for the development of the solution.

On this basis, and particularly considering that the information in question is already publicly available, it is the Data Protection Agency's opinion that the dataset can be shared on sprogteknologi.dk with reference to Article 6, paragraph 1, letter e, and Article 9, paragraph 2, letter g of the General Data Protection Regulation, with reference to the Public Access to Information and Administration Act as a supplementary legal basis.

****3.4.2. Fundamental Principles****

The sharing of the dataset, including particularly its publication, also raises a number of considerations in relation to the fundamental principles in Article 5 of the General Data Protection Regulation. This concerns, among other things, the issue of data minimization, cf. Article 5, paragraph 1, letter c, the accuracy of the information, cf. Article 5, paragraph 1, letter d, as well as the legality of the information, cf. Article 5, paragraph 1, letter a.

****Data Minimization****

According to the Data Protection Agency's guidelines for the use of personal data in the development and testing of IT systems, the following is stated, among other things: “It may, depending on the circumstances, be well-founded and necessary to use personal data in the development and testing of IT systems. For example, it would be acceptable to use personal data in connection with final tests of

integrations to other (external) IT systems. It would also be acceptable to use personal data in cases where it is associated with significant difficulties to create accurate (anonymized) test data, particularly because it can be challenging to reflect all the errors and irregularities that may occur in a production environment. Furthermore, it may be well-founded to use a limited number of personal data in connection with troubleshooting and debugging.

The use of personal data for testing purposes can therefore occur to a limited extent. The closer you (as a business or as an authority) get to the production phase, the more well-founded it may be that you use (multiple) production data – including personal data. [Sometimes] it may even be indicative of a lack of security to put a system into production without having tested it with production data – including personal data – first. [...]

****Data Minimization****

Personal data must be adequate, relevant, and limited to what is necessary in relation to the purposes for which they are processed. This means, among other things, that even where it is well-founded and necessary to use personal data in connection with testing IT systems, you must not use more personal data for testing purposes than is necessary to achieve the testing objective. It is also important that you do not use personal data to perform tests if the same tests could be conducted without using (real) personal data.”

Similarly, it is stated in the Data Protection Agency's guidance on public authorities' use of artificial intelligence (Before You Get Started), p. 10: “The proportionality assessment then entails that you must consider how the AI solution can be developed, trained, and operated using the fewest possible personal data – and if possible, entirely without personal data. The data protection rules do not, as mentioned, contain an actual prohibition against processing personal data in connection with the development and testing of new technological solutions, but the starting point is that anonymized data should be used to the greatest extent possible.”

In an AI context, there are several techniques that can be used to process fewer personal data. This includes, among other things, the use of synthetic data and federated learning. When developing an AI solution, you should consider the use of such techniques already in the design phase. You must make efforts to ensure that you process the least possible information when designing and developing the solution. There may be legitimate reasons to deviate from this starting point, but you must describe why the starting point is deviated from. The justification must, among other things, describe why it is not possible to use synthetic or anonymized data. A justification could, for example, be that the construction of suitable synthetic test data is impossible, or that without the use of personal data, there will be a risk that the forthcoming solution will generate incorrect outputs subsequently. Conversely, any costs associated with developing, for example, synthetic data cannot in themselves justify that the starting point can be deviated from. If you determine that this is not possible, as personal data is necessary for the development of the AI solution, you must, as a clear starting point, only use pseudonymized information.

According to the Data Protection Authority's view, this also applies in this situation where Sønderborg Municipality wishes to share the dataset that is, among other things, the basis for the development of the municipality's own models. Therefore, the municipality must, to the greatest extent possible, anonymize the information contained in the dataset before sharing, unless there are legitimate reasons for using personal data for the development of future models and therefore must be included in the dataset. In such cases, the municipality should consider whether it is possible to pseudonymize the information without compromising the purpose of sharing the dataset.

Correctness

If it is necessary for the dataset in question to contain personal data for the development of future models, it should be considered, according to the Data Protection Authority's view, to what extent the municipality is obliged to take steps to ensure that the information is up-to-date. Information collected from the internet can naturally become outdated as the original information on the internet changes or is removed. According to the Data Protection Authority's view, Sønderborg Municipality – to adequately

account for the principle of correctness – must, in connection with the sharing of the dataset, including particularly its publication, declare when the personal data in question was collected. It is then up to the company or authority that wishes to use the dataset to assess whether the intended use complies with data protection regulations, for example, whether it is lawful and accurate to develop a new AI model based on a dataset from that specific date.

Legality

Finally, information found on the internet may have been published unintentionally. This may involve breaches of personal data security caused by malicious actors or mere human errors that result in the information being published (leaked). According to the Data Protection Authority's view, the principle of legality in Article 5(1)(a) of the Regulation implies that companies and authorities cannot lawfully process information that has been unlawfully collected as part of the development and operation of IT systems. This may be the case, among other things, if the information has been published as a result of a breach of personal data security. Therefore, Sønderborg Municipality must, as part of, among other things, sharing the dataset, ensure that there is no information that is the result of unintentional publication. This can practically be done by ensuring:

That the municipality cannot ascertain from publicly available sources that the information originates from an unintentional publication, and

That the municipality cannot ascertain based on decisions from authorities, etc., that the information has been unlawfully published.

This does not mean that the municipality is obliged to review each individual piece of information to ensure that the information has been lawfully published, nor is the municipality obliged to verify whether the company or authority had the necessary legal basis to publish the information. Instead, the municipality must, at a more general level, ascertain that, as part of the establishment of the dataset, data has not been obtained from sources that are the result of unintentional or unlawful publication.

[1] Background, What is sprogteknologi.dk? <https://sprogteknologi.dk/pages/baggrund>

[2] The impact assessment can be accessed here: <https://ai-aktindsigt.dk/wp-content/uploads/2023/05/Til-offentliggørelse-DPIA-Signaturprojekt-AI-aktindsigt.pdf>

[3] The Data Protection Authority's statement to Copenhagen Municipality of November 17, 2023, j.nr. 2023-212-0015: <https://www.datatilsynet.dk/afgoerelser/afgoerelser/2023/nov/udtalelse-om-behandling-af-grundlag-til-udvikling-og-drift-af-ai-loesning-inden-for-sundheds-og-omsorgsomraadet>

[4] Kristian Korfits Nielsen and Anders Lotterup, The Data Protection Regulation and the Data Protection Act with comments, 2020, particularly p. 179f.

[5] See in the same direction Hanne Marie Motzfeldt and Azad Taheri Abkenar, Digital Administration, 2019, p. 76 with further references to the administrative law literature; Nikolaj Aarø-Hansen in Niels Fenger (ed.): Administrative Law, 2018, p. 626.

[6] The Data Protection Authority, Test Data – Use of Personal Data in the Development and Testing of IT Systems: <https://www.datatilsynet.dk/hvad-siger-reglerne/vejledning/sikkerhed/testdata-brug-af-personoplysninger-ved-udvikling-og-test-af-it-systemer>